How Reliable Is Explainable AI for Plant Disease Recognition? A Robustness Evaluation under Real-World Image Transformations

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Abstract. Deep neural networks achieve near-perfect accuracy on benchmark plant-disease image datasets, and post-hoc explanation methods such as Class Activation Mapping (CAM) are increasingly used to justify their predictions to agronomists. Yet field images are rarely pristine: they are blurred, rotated, unevenly lit, and noisy. We ask whether the *explanations* produced for such models remain reliable when the input undergoes realistic, label-preserving transformations. We introduce a reproducible benchmarking framework that couples a controlled image perturbation protocol (Gaussian blur, rotation, brightness scaling and additive noise) with two complementary axes of evaluation: prediction consistency and explanation stability. Using the unaugmented, leaf-split PlantVillage colour dataset and three ImageNet-pretrained backbones (ResNet-50 and EfficientNet-B0 for the explanation analysis; ViT-B/16 for classification context), we evaluate the CAM family (Grad-CAM, Grad-CAM++, Score-CAM, Eigen-CAM, Layer-CAM), quantifying explanation stability with Pearson correlation, SSIM and top-k IoU between the (geometrically aligned) original and perturbed saliency maps, and complementing it with deletion, insertion and average-drop faithfulness metrics. Statistical significance is assessed with bootstrap confidence intervals and Holm-corrected paired tests. Our study yields four findings that are hidden by accuracy alone: (i) explanation stability varies strongly by perturbation — brightness is almost inert whereas blur and noise are highly disruptive; (ii) Layer-CAM, Grad-CAM++ and Eigen-CAM are the most stable and Grad-CAM reliably the least (Holm-corrected, pooled and in 16/24 per-transform tests), yet Grad-CAM is the most faithful — an explicit stability–faithfulness trade-off; (iii) sensitivity is architecture-dependent, with ResNet-50 most vulnerable to noise and EfficientNet-B0 most vulnerable to blur; and (iv) explanation fragility gets *worse*, not better, as the model approaches deployment accuracy: on a fully fine-tuned 98.9%-accurate model the gradient/score methods (Grad-CAM, Score-CAM) degrade sharply while element-wise and principal-component maps stay robust. We frame deployment implications as hypotheses for field validation and release the full protocol, code and raw measurements to support reproducible robustness auditing of agricultural XAI.

Keywords: Explainable AI · Robustness · Class Activation Mapping · Plant disease recognition · Trustworthy machine learning · Benchmarking

1 Introduction

Automated plant disease recognition from leaf images is one of the most mature applications of deep learning in precision agriculture. Convolutional and, more recently, transformer architectures routinely exceed 99% accuracy on curated benchmarks and are being embedded in mobile diagnostic tools used by growers and extension agents [14,6,21]. Because an incorrect diagnosis can trigger unnecessary pesticide application or crop loss, the deployment of such models in agronomy is increasingly gated on *trust*: a practitioner must be able to see *why* the model reached its conclusion. Explainable AI (XAI), and in particular saliency methods from the Class Activation Mapping (CAM) family, has become the default vehicle for this transparency, highlighting the leaf regions deemed responsible for a prediction [24,19,3]. The need for interpretable and trustworthy models is not unique to agriculture: it recurs wherever machine learning informs high-stakes decisions, from interpretable strength prediction in materials science [17] to machine-learning-based medical image classification [16], where an opaque prediction is difficult to act upon.

A saliency map is only useful if it is *reliable*, and two failure modes matter: the prediction may change under innocuous input variation, or — more insidiously — the prediction may stay correct while the *explanation* shifts to unrelated regions, silently invalidating the justification shown to the user. Interpretations are known to be fragile to perturbation in generic vision settings [7,2,1], yet the plant-disease community has largely evaluated XAI qualitatively, on clean laboratory images, without a statistically grounded protocol — even though field acquisitions are routinely degraded by blur, tilt, uneven lighting and noise. Whether the explanations underpinning agronomic trust survive these everyday transformations is, to our knowledge, open.

Research questions. We study two questions. *RQ1 (prediction consistency):* how much do label-preserving image transformations degrade classification, and does this depend on the backbone? *RQ2 (explanation stability):* do CAM explanations remain spatially consistent under the same transformations, how do CAM variants compare, and does stability align with faithfulness?

Contributions.

- A *reproducible image-perturbation protocol* for auditing agricultural XAI, comprising geometric, photometric and noise-based transformations with a geometry-aware alignment step that makes original and perturbed saliency maps directly comparable.
- A *benchmarking framework* that jointly measures prediction consistency and explanation stability (Pearson, SSIM, top-k IoU) alongside established faithfulness metrics (deletion, insertion, average drop), with bootstrap confidence intervals and Holm-corrected significance testing.
- An *empirical study* on the leaf-split PlantVillage colour dataset — classification across three backbones, and the explanation analysis on the two CNNs (four CAM methods on both, Score-CAM on EfficientNet-B0) — reporting *honest* results including cases where methods lose. It surfaces effects that accuracy alone conceals (a stability–faithfulness trade-off and architecture-dependent sensitivity), and adds a fully fine-tuned validation.
- Two diagnostic controls on the deployment-accuracy model: a Score-CAM weighting collapse (with a candidate fix) and evidence that the most-recommended CAM variants become near-class-agnostic — so stability must be read jointly with faithfulness.
- A public release of code, the exact evaluation manifests and all raw per-image measurements to enable exact reproduction and extension.

The remainder of the paper is organised as follows. Section 2 reviews architectures, CAM methods and XAI evaluation. Section 3 formalises the problem and Section 4 details the framework. Section 5 reports experiments and discussion, and Section 6 concludes.

2 Related Work

From descriptors to deep architectures. Deep learning displaced handcrafted descriptors for plant-disease recognition: residual networks [10], EfficientNet's compound scaling [20] and the Vision Transformer [5] all report $> 99\%$ accuracy on PlantVillage and related corpora [14,6,21], motivating a shift of attention from raw accuracy to reliability and explanation quality.

CAM-family explanations. CAM [24] localises class evidence via global-average-pooled feature maps. Grad-CAM generalises it to arbitrary CNNs using gradients flowing into the last convolutional layer [19]; Grad-CAM++ refines the weighting with higher-order gradient terms for better multi-instance localisation [4]; Score-CAM removes the gradient dependency, weighting each activation map by the confidence it induces through a forward pass [22]; Eigen-CAM takes the leading principal component of the activation maps and is gradient-free [15]; and Layer-CAM re-weights activations element-wise to yield sharper, hierarchical maps [13]. Perturbation- and gradient-based attributions such as RISE [18] form a complementary family; we focus on CAM methods because they dominate agricultural XAI practice.

Evaluating explanations. Explanation quality is assessed along *faithfulness* (deletion/insertion curves [18], average drop [4]), *localisation*, and *sanity* — sensitivity to model and data [1]; a separate line argues that *robustness* (similar inputs → similar explanations) is itself a desideratum that many attributions violate [2,7], and toolkits such as Quantus [11] consolidate these metrics. SSIM [23] naturally compares two saliency maps.

Positioning. Most agricultural XAI studies remain qualitative and use controlled imagery, leaving robustness under realistic acquisition largely unquantified [9,3]. We fill this gap with a statistically grounded framework that measures *both* prediction consistency and explanation stability under a reproducible transformation protocol, reporting faithfulness jointly so trade-offs are explicit.

3 Problem Formulation

Let $f : \mathcal{X} \to \mathbb{R}^C$ be a trained classifier mapping an image $x \in \mathcal{X} \subset \mathbb{R}^{3\times H\times W}$ to logits over C classes, with predicted label $\hat{y}(x) = \arg\max_c f_c(x)$. A CAM explainer Φ produces a saliency map $\Phi(f, x, c) \in [0, 1]^{H\times W}$ for target class c; we always explain the class predicted on the *original* image, $c = \hat{y}(x)$, so that stability is measured for a fixed concept. We note that the methods differ in how strongly they depend on c: Grad-CAM, Grad-CAM++, Layer-CAM and Score-CAM are class-conditional (through the class logit or its gradients), whereas Eigen-CAM explains the leading principal component of the activation maps and is effectively class-agnostic. For Eigen-CAM, "stability of the explanation for class c" therefore reduces to stability of the dominant activation subspace; we retain it as a widely used gradient-free baseline and interpret its scores with this caveat rather than as a like-for-like class-conditional comparison.

Let $\mathcal{T} = \{T_1, \ldots, T_m\}$ be a set of label-preserving transformations $T : \mathcal{X} \to \mathcal{X}$ (Section 4). For a transformation T we define two reliability notions.

Prediction consistency. For an evaluation set S,

$$\mathrm{PC}(T) = \frac{1}{|S|} \sum_{x\in S} \mathbb{1}[\hat{y}(T(x)) = \hat{y}(x)], \tag{1}$$

where $\mathbb{1}[\cdot]$ is the indicator function. We also report the drop in macro-averaged F_1 between the clean and transformed sets.

Explanation stability. A geometric transformation moves salient content, so a naive pixel-wise comparison of $\Phi(f, x, \hat{y})$ and $\Phi(f, T(x), \hat{y})$ would conflate genuine instability with expected motion. We therefore align the original map into the transformed frame with the same geometric operation, $A_T(\cdot)$ ($A_T = T$ for rotation, identity for photometric transforms), and restrict comparison to the valid region M_T (pixels defined after A_T). Writing $u = A_T(\Phi(f, x, \hat{y}))$ and $v = \Phi(f, T(x), \hat{y})$, stability is quantified by three metrics over M_T: Pearson correlation $r(u, v)$; structural similarity $\mathrm{SSIM}(u, v)$; and the intersection-over-union of the top-k salient pixels,

$$\mathrm{IoU}_k(u, v) = \frac{|B_k(u) \cap B_k(v)|}{|B_k(u) \cup B_k(v)|}, \qquad B_k(w) = \{p : w_p \geq Q_{1-k}(w)\}, \tag{2}$$

with Q_{1-k} the $(1-k)$ quantile. We report $k = 0.2$ (top quintile); a sensitivity check over $k \in \{0.1, 0.2, 0.3\}$ leaves the IoU-based ordering unchanged (Eigen-CAM > Layer-CAM > Grad-CAM++ > Grad-CAM at every k on EfficientNet-B0; this is the IoU ordering, cf. the Pearson-based ranking in Table 2), so the threshold does not drive our conclusions. Higher values indicate more stable explanations. Perfect stability gives $r = \mathrm{SSIM} = \mathrm{IoU}_k = 1$.

Faithfulness. Independently of transformations, we assess how well Φ reflects f on the clean image via deletion and insertion AUC [18] and average drop [4] (Section 4). Reliability requires stability *and* faithfulness; the framework reports both so trade-offs are visible.

4 Evaluation Framework

Dataset. We use the PlantVillage colour configuration obtained programmatically from the official `mohanty/PlantVillage` Hugging Face repository. We adopt the repository's *leaf-based* train/test split, which groups all views of a physical leaf on one side of the split and thereby prevents leakage between near-duplicate images. To keep the study executable at CPU scale we draw a class-balanced subset with a fixed seed: 35 training and 25 test images per class across all 38 {crop, condition} classes (1,330 train / 950 test images). Images are resized to 160×160 (CNNs) and normalised with ImageNet statistics.

Backbones and training. We evaluate three ImageNet-pretrained backbones: ResNet-50 [10], EfficientNet-B0 [20] and ViT-B/16 [5] (ViT uses its native 224×224 input). Given the memory budget we adopt *linear probing*: backbone weights are frozen and a linear classification head is trained on the pooled features (global-average-pooled last convolutional map for CNNs; the class token for ViT) using Adam. Freezing the backbone does not affect CAM computation, which back-propagates through the frozen convolutional stack to the target layer.

Transformation protocol. Four label-preserving transformations model common field degradations, applied to the $[0, 1]$ image before normalisation: *Gaussian blur* (9×9 kernel, $\sigma = 2$), *rotation* ($+15°$, bilinear), *brightness* scaling ($\times 1.3$, clipped) and additive *Gaussian noise* ($\sigma = 0.08$). The identity transform serves as reference.

CAM methods. We evaluate Grad-CAM, Grad-CAM++, Eigen-CAM and Layer-CAM on both CNN backbones, and additionally Score-CAM on EfficientNet-B0. Score-CAM performs one forward pass per activation channel (1,280 for EfficientNet-B0, 2,048 for ResNet-50); on CPU this is roughly 50–90$\times$ slower than the gradient methods, so we restrict it to EfficientNet-B0 on a 12-image subset and report it with its own sample size. All maps are computed with a public CAM library [8] on the last convolutional block.

Evaluation sets. Prediction consistency (Eq. 1) is measured on a 190-image set (5 per class). Explanation stability and faithfulness are measured on a 38-image set (one image per class that is correctly classified on the *clean* input, shared across backbones); this smaller set reflects the cost of computing five CAMs under five conditions per image and is reflected in the reported confidence intervals. A transformation may flip the prediction; because we always explain the clean predicted class c, such an image is *not* removed from the stability computation — the case in which the prediction changes (or, worse, stays correct while the map migrates) is exactly the failure mode we wish to capture. Prediction-level effects are reported separately in Section 5.2.

Faithfulness computation. Deletion progressively zeroes the most salient pixels (low AUC is better); insertion progressively reveals them from a blurred baseline (high AUC is better); both use 20 steps. Average drop is $\max(0, p_{\text{full}} - p_{\text{mask}})/p_{\text{full}} \times 100$, where p_{mask} uses the CAM as a soft mask (lower is better).

Statistics. We compute bootstrap 95% CIs (2,000 resamples) for all means; to respect the page limit they are printed in Tables 2 and 4 and in the released records for the rest. Pairwise comparisons use per-image mean Pearson stability with Wilcoxon signed-rank tests and Holm correction within each backbone [12].

5 Experiments and Results

Implementation. All experiments run on CPU (4 cores, PyTorch). Seeds are fixed (42) throughout; the released code reproduces every table and figure from the saved per-image records.

5.1 Classification performance

Under linear probing on the 38-class task (950 test images), clean accuracy / macro-F_1 is 0.868/0.867 (ResNet-50), 0.883/0.882 (EfficientNet-B0) and 0.919/0.918 (ViT-B/16). These are below the near-saturated numbers usually quoted for PlantVillage — because we freeze the backbone and subsample — but are more than sufficient as a substrate for a relative robustness study (§5.6 revisits this with a fully fine-tuned 98.9% model).

5.2 RQ1: Prediction consistency under transformation

Table 1 quantifies prediction robustness. Brightness scaling is almost inert (prediction consistency PC = 0.95 for both backbones; macro-F_1 essentially unchanged). Rotation causes a moderate drop (PC = 0.83 for ResNet-50, 0.77 for EfficientNet-B0). Blur and noise are the most damaging, but the ranking is architecture-dependent: ResNet-50 is most harmed by noise (PC = 0.53, macro-F_1 -0.37) whereas EfficientNet-B0 is most harmed by blur (PC = 0.53, macro-F_1 -0.38). Thus even at the prediction level, robustness is not a property of the task alone but of the model, a distinction that a single clean-accuracy number hides.

Table 1. Prediction consistency (PC, Eq. 1) and macro-F_1 drop relative to clean, per transformation (190-image set).

	ResNet-50		EfficientNet-B0	
Transform	PC	ΔF_1	PC	ΔF_1
Blur	0.621	−0.283	0.532	−0.375
Rotate	0.832	−0.086	0.774	−0.127
Brightness	0.947	+0.008	0.947	−0.007
Noise	0.532	−0.372	0.695	−0.212

Table 2. Explanation stability averaged over the four transformations (mean with bootstrap 95% CI; higher is better). n=38 except Score-CAM (n=12). The wide, overlapping CIs are consistent with the significance analysis: methods are *ordered* but individual per-transform differences are not powered at n=38.

Backbone	Method	Pearson r	SSIM	$\text{IoU}_{0.2}$
ResNet-50	Grad-CAM	0.769 [0.722,0.813]	0.673 [0.644,0.700]	0.561 [0.518,0.601]
	Grad-CAM++	0.879 [0.853,0.900]	0.806 [0.789,0.821]	0.632 [0.593,0.670]
	Eigen-CAM	0.873 [0.832,0.907]	0.789 [0.757,0.817]	0.697 [0.657,0.737]
	Layer-CAM	**0.882** [0.857,0.904]	**0.809** [0.793,0.824]	0.637 [0.598,0.674]
EfficientNet-B0	Grad-CAM	0.771 [0.704,0.831]	0.788 [0.758,0.815]	0.604 [0.552,0.653]
	Grad-CAM++	0.872 [0.828,0.909]	0.796 [0.771,0.818]	0.689 [0.638,0.733]
	Eigen-CAM	0.824 [0.738,0.897]	**0.802** [0.754,0.846]	**0.724** [0.659,0.782]
	Layer-CAM	**0.873** [0.833,0.909]	0.799 [0.774,0.822]	0.690 [0.643,0.731]
	Score-CAM	0.761 [0.672,0.842]	0.727 [0.675,0.783]	0.596 [0.500,0.692]

5.3 RQ2: Explanation stability

Table 2 summarises stability, averaged over transforms (per-transform values in Appendix A). The picture mirrors RQ1: brightness leaves explanations almost untouched ($r > 0.95$), while blur and noise dominate instability. **Layer-CAM** and **Grad-CAM++** are the most stable ($r = 0.88/0.87$), **Grad-CAM** consistently the least ($r = 0.77$). Eigen-CAM is competitive on ResNet-50 ($r = 0.87$) but degrades under rotation on EfficientNet-B0 ($r = 0.67$); Score-CAM is already the least stable under blur ($r = 0.39$, wide CI, small sample), foreshadowing its fine-tuned collapse in §5.6 (Eigen-CAM, by contrast, remains the *most* stable there).

Significance. On the pooled score (per-image Pearson averaged over the four transformations), Holm-corrected Wilcoxon tests find Grad-CAM significantly less stable than Grad-CAM++, Eigen-CAM and Layer-CAM on ResNet-50 (all $p_{\text{Holm}} < 10^{-3}$) and than Grad-CAM++ and Layer-CAM on EfficientNet-B0 ($p_{\text{Holm}} = 0.014$). The effect also holds *per transformation*: of the 24 Grad-CAM-vs-other comparisons (2 backbones × 4 transforms × 3 pairs, Score-CAM excluded; full table in Appendix A), Grad-CAM is significantly less stable in 16 ($p_{\text{Holm}} < 0.05$), most robustly under noise and blur ($p_{\text{Holm}} < 10^{-3}$); it is not sig-

Table 3. Faithfulness on clean images (mean; 95% CIs in the released records). Deletion↓, Insertion↑, Average drop (%)↓. n=38 except Score-CAM (n=12).

Backbone	Method	Deletion	Insertion	Avg. drop
ResNet-50	Grad-CAM	0.363	0.559	**35.6**
	Grad-CAM++	0.393	0.558	36.9
	Eigen-CAM	0.399	**0.562**	57.8
	Layer-CAM	0.391	0.559	36.9
EfficientNet-B0	Grad-CAM	0.195	0.626	**19.9**
	Grad-CAM++	0.187	0.621	45.6
	Eigen-CAM	0.212	0.611	66.5
	Layer-CAM	**0.183**	0.622	45.1
	Score-CAM	0.214	**0.701**	37.9

nificant only under brightness (where all methods are near-ceiling) and for a few rotation pairs (per-transform table in the released records). Differences among the top three methods are small — the near-identical Grad-CAM++/Layer-CAM pair can reach significance on negligible (< 0.01) gaps, which we do not interpret substantively — so we treat Layer-CAM, Grad-CAM++ and Eigen-CAM as jointly most stable and Grad-CAM as reliably least.

5.4 Faithfulness and the stability–faithfulness trade-off

Table 3 reports faithfulness. Strikingly, the ranking partly inverts that of stability. On EfficientNet-B0, **Grad-CAM** attains by far the best average drop (19.9 vs. 45–66 for the others) and a competitive deletion AUC, i.e. the very method that is *least stable* is among the *most faithful* to the model on clean inputs. Score-CAM achieves the best insertion AUC (0.70). This trade-off has a practical reading: a method that hugs the model's decision surface tightly (faithful) can also track spurious, perturbation-induced fluctuations (unstable), whereas smoother maps (Layer-CAM, Grad-CAM++) are more stable but less sharply tied to the logit. Reliability assessments that optimise only one axis are therefore misleading; both must be reported.

5.5 Qualitative analysis

Figure 1 shows a low-stability case: for a diseased corn leaf (common rust) whose prediction is preserved, the heatmaps migrate substantially under blur, rotation and noise — drifting off the visible rust pustules — while remaining nearly fixed under brightness. This is exactly the insidious failure mode motivating the study: the label is right, but the explanation no longer coincides with the disease evidence it highlighted on the clean image. High-stability-quantile cases (released records) instead show near-identical maps across all conditions, so instability is concentrated in a subset of images and perturbations, not uniform.

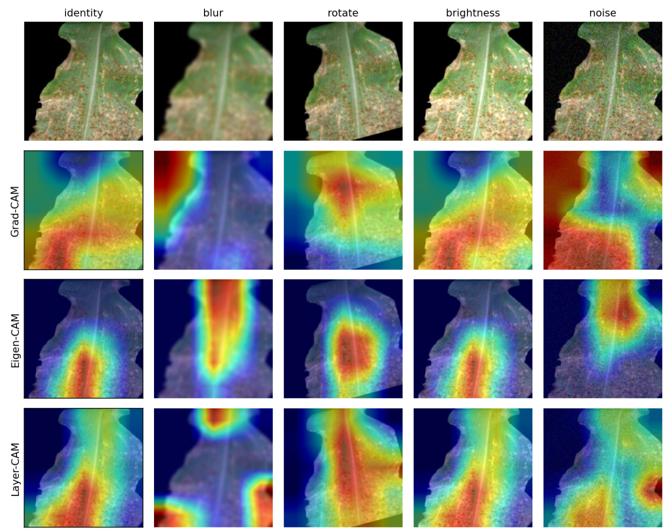

Fig. 1. A low-stability example on a *diseased* leaf (EfficientNet-B0, corn common rust; rust pustules visible across the lamina). Rows: representative CAM methods; columns: identity and the four transforms (Grad-CAM++ and Score-CAM behave similarly; full 5-method montage in the released records). The prediction is preserved, yet the heatmaps migrate markedly under blur, rotation and noise — drifting off the lesion-bearing regions — while brightness is benign.

5.6 Fine-tuned validation and two controls

To test whether the frozen-backbone findings persist once features specialise to disease cues, we fully fine-tuned EfficientNet-B0 end-to-end; it reaches **98.9%** accuracy / macro-F_1 — the near-saturated regime of deployed models, up from 88.3% for the linear probe. Re-running the protocol on the *same* 38 images (paired comparison, Table 4) shows the stable-method ranking transfers — Eigen-CAM, Layer-CAM and Grad-CAM++ stay high ($r \approx 0.79$–0.83) — while Grad-CAM roughly halves ($0.77 \rightarrow 0.39$; on a larger n=100 set it falls to 0.20). Fine-tuning thus makes gradient-weighted maps more brittle, not less: explanation fragility *worsens* as the model approaches deployment accuracy.
Control 1 — the Score-CAM collapse is a property of the published method, not a bug. On the fine-tuned model Score-CAM falls to $r = 0.04$ (n=100). The cause is the weighting that the original Score-CAM formulation *specifies*: channels are weighted by a softmax over their per-channel target *logits*. On the confident model this distribution peaks sharply — its entropy drops from 6.25 to 2.16 (max 7.15) and a single channel captures 53% of the weight (vs. 1% frozen) — so the map collapses onto one volatile channel. As a *proposed modification*, replacing the softmax with a bounded min–max channel weighting more than doubles stability on a small diagnostic set (4 easier images, hence a higher 0.28 baseline than the n=100 headline of 0.04): mean Pearson rises

Table 4. Frozen (linear-probe) vs. fully fine-tuned EfficientNet-B0 on the *same* 38 images: explanation stability (Pearson r, mean [95% CI]) and class-sensitivity (higher = more class-conditional; see Control 2). Score-CAM is treated separately in Control 1. The three near-class-agnostic methods are the three that stay stable; the strongly class-conditional method (Grad-CAM) collapses.

Method	Frozen r	Fine-tuned r (98.9% acc.)	Class-sensitivity
Grad-CAM	0.771 [0.704,0.831]	0.392 [0.313,0.475]	0.97
Grad-CAM++	0.872 [0.828,0.909]	0.808 [0.777,0.840]	0.03
Eigen-CAM	0.824 [0.738,0.897]	**0.828** [0.795,0.860]	0.00
Layer-CAM	**0.873** [0.833,0.909]	0.793 [0.758,0.828]	0.02

$0.28 \rightarrow 0.64$. Since the target is already the pre-softmax logit, this is distinct from classic softmax-gradient saturation; a full-scale evaluation is future work.

Control 2 — stability can be inflated by class-insensitivity. The fine-tuned stability "winners" are exactly the methods that barely depend on the predicted class. Measuring class-sensitivity as $1 - r$ between a method's map for the predicted class and for a single random non-predicted class (one draw per image, seed 42; Table 4, last column), the three methods with near-zero class-sensitivity (Eigen-CAM 0.00, Grad-CAM++ 0.03, Layer-CAM 0.02) are precisely the three that stay stable, while the one strongly class-conditional method (Grad-CAM, 0.97) is the one that collapses. This is not an artefact of Grad-CAM's own instability: a self-consistency control — $1 - r$ between a method's map on the image and on a jittered copy of the *same* class (σ=0.02 Gaussian) — is ≈ 0.01 for *every* method, Grad-CAM included. Grad-CAM's maps are thus self-consistent under input noise yet change almost completely when the target class changes (0.01 vs. 0.97), so its high class-sensitivity is genuine class-discrimination; the other methods' near-zero values are genuine class-*agnosticism*, not measurement noise. A map that scarcely moves when the target class changes also has little reason to move under perturbation, so *stability alone can be inflated by uninformative explanations* — which is exactly why our framework reads stability jointly with faithfulness (Table 5) rather than in isolation.

Beyond our metric, this is a finding about the *methods*. Grad-CAM++ and Layer-CAM are class-conditional *by construction*, yet on the model closest to deployment their maps barely change when a random target class is substituted — a failure of the model-and-data sensitivity that saliency sanity checks demand [1]. In other words, at deployment-level accuracy the CAM variants most recommended in the agricultural literature can produce near-class-agnostic explanations; accuracy conceals this, and stability alone would have rewarded it.

Faithfulness under perturbation. To make the trade-off axis like-for-like, we measure faithfulness on perturbed inputs as well, reporting each deletion/insertion AUC as a *fraction of the input's own unmasked confidence* so the comparison is not driven by the base-rate confidence drop already reported in §5.2 (on a

Table 5. Fine-tuned EfficientNet-B0: confidence-normalised deletion/insertion AUC on clean vs. perturbed inputs (fraction of the input's own unmasked probability; mean over four transforms, n=24). Clean columns read as usual (↓/↑). Under perturbation, values > 1 (a masked input beating the full input) are a degeneracy signal — the saliency ordering has become anti-correlated with the model's evidence, *not* a gain — so the arrows do not apply to the perturbed columns.

Method	Del. clean↓	Del. pert.	Ins. clean↑	Ins. pert.
Grad-CAM	0.291	0.841	0.386	2.107
Grad-CAM++	0.291	1.272	0.392	1.560
Eigen-CAM	0.293	1.187	0.432	1.548
Layer-CAM	0.293	1.354	0.393	1.661

cost-bounded n=24 subset, as each curve needs ∼30 forward passes per image, method and condition). On clean inputs the corrected metrics behave as expected for a 98.9% model — insertion (0.39–0.43) exceeds deletion (0.29). *Even after normalisation*, faithfulness degrades sharply under perturbation: normalised deletion AUC rises from 0.29 to 0.84–1.35 (2.9–4.6×), with values > 1 signalling a degenerate regime where removing the salient pixels no longer collapses confidence (see caption). Faithfulness is thus genuinely perturbation-dependent, not a base-rate artefact, and should be assessed on the decision-relevant perturbed inputs. Notably, Grad-CAM is the *only* method whose perturbed deletion stays below 1 (0.84 vs. 1.19–1.35), so it remains the most faithful under perturbation — extending the trade-off from clean to the decision-relevant perturbed regime.

5.7 Discussion

Three factors govern explanation reliability. *Perturbation type* dominates: brightness is harmless, whereas blur and noise — which alter local texture, the cue for lesion recognition — are highly disruptive. *Method* matters next: element-wise maps (Layer-CAM, Grad-CAM++) are most stable but stability does not imply faithfulness (Grad-CAM is least stable yet most faithful, and §5.6 shows stability can even be inflated by class-insensitivity). *Architecture* modulates both, with ResNet-50 and EfficientNet-B0 showing opposite blur/noise sensitivities. As *hypotheses for field validation*, one might (i) prefer Layer-CAM or Grad-CAM++ when robustness to acquisition noise matters — but only after verifying they pass a class-sensitivity check on the deployed model, since (Control 2) their stability can otherwise reflect class-agnostic maps; (ii) always report stability *and* faithfulness; and (iii) deblur/denoise before trusting a saliency map.

A validity caveat on lab-vs-field. PlantVillage is a controlled, uniform-background corpus and our degradations are *simulated*; robustness to synthetic degradations on laboratory images is not the same as robustness on field acquisitions, where background clutter, occlusion and natural lighting can dominate saliency on their own. The recommendations above are therefore hypotheses to be tested on field data — the *protocol* (alignment, metrics, statistics) is what we claim transfers, not the specific magnitudes.

Limitations. The main experiments use frozen ImageNet backbones with a trained linear head (87–92% accuracy); we validated the central conclusion on a fully fine-tuned 98.9% EfficientNet-B0 (Sec. 5.6), where the stable-method ranking transfers and the Grad-CAM/Score-CAM fragility is in fact accentuated, but a broader sweep of fine-tuned architectures remains future work. All explanation analysis is on CNNs; ViT-B/16 is reported for classification context only, and transformer-native attributions (e.g. attention rollout) are left to future work, so our stability conclusions are CNN-specific. The controls and the perturbed-faithfulness analysis are also scope-bounded: all are single-architecture (EfficientNet-B0) and small-n (24–38), and the Score-CAM diagnosis is tied to one widely used library implementation of the published method (though the weighting it uses is the one the method specifies). Further bounds — a class-balanced subset, a 38-image frozen explanation set, single-severity transformations, and reduced Score-CAM sampling — widen confidence intervals but do not affect the protocol, alignment or statistics.

6 Conclusion and Future Work

We asked whether explainable AI for plant disease recognition is reliable when inputs undergo everyday transformations, and built a reproducible framework to answer it quantitatively. On the leaf-split PlantVillage colour dataset, we found that explanation stability is strongly perturbation-dependent (brightness benign; blur and noise damaging), that Grad-CAM is significantly less stable than the other CAM variants (all three on ResNet-50; Grad-CAM++ and Layer-CAM on EfficientNet-B0) while being the most faithful — an explicit stability–faithfulness trade-off — and that sensitivity is architecture-dependent; accuracy alone conceals all three. A fine-tuned 98.9%-accurate EfficientNet-B0 shows the *stability* half of these findings not only transfers but worsens (Sec. 5.6); two controls establish that Score-CAM's collapse is a diagnosable weighting artifact with a candidate fix, and that stability must be read jointly with faithfulness because it can otherwise be inflated by class-insensitive maps. Future work will extend the fine-tuned analysis to more architectures and to field-acquired imagery (e.g. Plant Pathology and citrus datasets), add multiple perturbation severities and combinations, cover transformer-native attributions, and study explanation-robust training objectives that regularise saliency maps to be invariant to label-preserving transforms.

Reproducibility. Code, evaluation manifests and all raw per-image measurements are released with the paper; every table and figure is regenerated from the saved records by the provided scripts.

Appendix A: per-transformation stability. Table 6 gives the per-transformation mean r (EfficientNet-B0, frozen) underlying Table 2 and the values quoted in

§5.3; ResNet-50 means and the Holm-corrected p-values behind the 16/24 count are in the released records.

Table 6. Per-transformation explanation stability (Pearson r, mean) for EfficientNet-B0 (frozen; n=38, 12 for Score-CAM).

Method	Blur	Rotate	Brightness	Noise
Grad-CAM	0.606	0.758	0.980	0.741
Grad-CAM++	0.780	0.821	0.983	0.902
Eigen-CAM	0.811	0.671	0.991	0.822
Layer-CAM	0.781	0.823	0.983	0.904
Score-CAM	0.387	0.834	0.953	0.870

References

1. Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M., Kim, B.: Sanity checks for saliency maps. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 31 (2018)
2. Alvarez-Melis, D., Jaakkola, T.S.: On the robustness of interpretability methods. arXiv preprint arXiv:1806.08049 (2018), iCML Workshop on Human Interpretability in Machine Learning (WHI)
3. Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., Chatila, R., Herrera, F.: Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. Information Fusion **58**, 82–115 (2020). https://doi.org/10.1016/j.inffus.2019.12.012
4. Chattopadhay, A., Sarkar, A., Howlader, P., Balasubramanian, V.N.: Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks. In: IEEE Winter Conference on Applications of Computer Vision (WACV). pp. 839–847 (2018)
5. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An image is worth 16x16 words: Transformers for image recognition at scale. In: International Conference on Learning Representations (ICLR) (2021)
6. Ferentinos, K.P.: Deep learning models for plant disease detection and diagnosis. Computers and Electronics in Agriculture **145**, 311–318 (2018). https://doi.org/10.1016/j.compag.2018.01.009
7. Ghorbani, A., Abid, A., Zou, J.: Interpretation of neural networks is fragile. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 33, pp. 3681–3688 (2019)
8. Gildenblat, J., contributors: PyTorch library for CAM methods. https://github.com/jacobgil/pytorch-grad-cam (2021)
9. Hamdaoui, W., Workneh, A.D., El Hilali Alaoui, A., Elmouhtadi, M.: Machine learning and explainable artificial intelligence for plant disease recognition: A systematic review of methods, datasets, and trustworthiness. Computers and Electrical Engineering **134**, 111130 (2026). https://doi.org/10.1016/j.compeleceng.2026.111130

10. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 770–778 (2016)
11. Hedström, A., Weber, L., Krakowczyk, D., Bareeva, D., Motzkus, F., Samek, W., Lapuschkin, S., Höhne, M.M.C.: Quantus: An explainable AI toolkit for responsible evaluation of neural network explanations and beyond. Journal of Machine Learning Research **24**(34), 1–11 (2023)
12. Holm, S.: A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics **6**(2), 65–70 (1979)
13. Jiang, P.T., Zhang, C.B., Hou, Q., Cheng, M.M., Wei, Y.: LayerCAM: Exploring hierarchical class activation maps for localization. IEEE Transactions on Image Processing **30**, 5875–5888 (2021). https://doi.org/10.1109/TIP.2021.3089943
14. Mohanty, S.P., Hughes, D.P., Salathé, M.: Using deep learning for image-based plant disease detection. Frontiers in Plant Science **7**, 1419 (2016). https://doi.org/10.3389/fpls.2016.01419
15. Muhammad, M.B., Yeasin, M.: Eigen-CAM: Class activation map using principal components. In: International Joint Conference on Neural Networks (IJCNN). pp. 1–7 (2020)
16. Nageshwaran, V.: Optimizing feature selection for brain tumor classification using a hybrid ACCSA and utilizing machine learning algorithms for classification. In: International Conference on Smart Electronics and Communication (ICOSEC) (2024)
17. Nageshwaran, V., Amritphale, S., Ezekiel, S.: Interpretable machine learning for one-part fly-ash/slag geopolymer strength prediction: Toward multifunctional binder design. Preprints (2026)
18. Petsiuk, V., Das, A., Saenko, K.: RISE: Randomized input sampling for explanation of black-box models. In: British Machine Vision Conference (BMVC) (2018)
19. Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-CAM: Visual explanations from deep networks via gradient-based localization. International Journal of Computer Vision **128**(2), 336–359 (2020). https://doi.org/10.1007/s11263-019-01228-7
20. Tan, M., Le, Q.V.: EfficientNet: Rethinking model scaling for convolutional neural networks. In: International Conference on Machine Learning (ICML). pp. 6105–6114 (2019)
21. Too, E.C., Yujian, L., Njuki, S., Yingchun, L.: A comparative study of fine-tuning deep learning models for plant disease identification. Computers and Electronics in Agriculture **161**, 272–279 (2019). https://doi.org/10.1016/j.compag.2018.03.032
22. Wang, H., Wang, Z., Du, M., Yang, F., Zhang, Z., Ding, S., Mardziel, P., Hu, X.: Score-CAM: Score-weighted visual explanations for convolutional neural networks. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). pp. 24–25 (2020)
23. Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: From error visibility to structural similarity. IEEE Transactions on Image Processing **13**(4), 600–612 (2004). https://doi.org/10.1109/TIP.2003.819861
24. Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., Torralba, A.: Learning deep features for discriminative localization. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2921–2929 (2016)